The Economics of Institutional Memory

Why Markets Record Possession Better Than Contribution

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November 2026

Series	PBS/MPS Foundations - Paper 2 of 6
Research status	Author synthesis and proposed economic research framework
Canonical question	When does institutional memory of contribution create more economic value than it costs to preserve?
Keywords	institutional memory, provenance, transaction costs, information asymmetry, incomplete contracts, hold-up, bargaining power, cumulative innovation, externalities

Abstract

Markets possess extraordinary memory for some kinds of economic facts. They preserve title, shares, balances, contracts, invoices, payments, liens, transfers, and prices with enough precision to support exchange among strangers across long periods of time. They are substantially less systematic at preserving the causal history of value creation: which people supplied the ideas, judgments, data, refinements, experiments, heuristics, relationships, and complementary efforts from which the final asset or productive capability emerged. The asymmetry is not mysterious. Possession is often a low-dimensional state that institutions must record in order to transfer rights. Contribution is typically a high-dimensional history that is costly to identify, verify, decompose, monitor, contest, and maintain.

This paper develops an economic theory of that asymmetry. It treats provenance as a form of institutional memory and asks when preserving contribution information improves economic coordination enough to justify its cost. The analysis integrates information asymmetry, transaction-cost economics, incomplete contracting, principal-agent problems, asset specificity and hold-up, bargaining power, path dependence, cumulative innovation, and positive externalities. It distinguishes the evidentiary value of provenance from any later legal or moral rule that might convert evidence into ownership, payment, or another entitlement.

The central proposition is deliberately conditional: more institutional memory is not always economically desirable. Provenance is worthwhile only when the expected reduction in information, verification, contracting, bargaining, monitoring, dispute, and misallocation costs exceeds the full cost of recording, validating, attributing, contesting, governing, protecting, and administering the information. This yields a testable research program. If credible, portable contribution histories improve investment, licensing, labor mobility, innovation, price discovery, or dispute resolution in some domains but not others, the economically appropriate boundary of provenance can be discovered rather than assumed.

I. The Forgotten Production History

Every market transaction contains a compressed history. A share certificate records a current ownership interest, not every contribution that made the enterprise valuable. A patent records a legally recognized invention and chain of rights, not the entire web of prior knowledge, experimentation, engineering support, institutional infrastructure, failed attempts, and complementary improvements that made commercialization possible. A wage records compensation under an employment relationship, not necessarily the long-run economic effect of each idea or process refinement produced within it. A price records an exchange outcome, not a causal decomposition of the value exchanged.

This compression is economically useful. Market institutions cannot operate if every transaction requires a reconstruction of the whole history of production. Transferability depends on abstraction. A buyer of land needs a reliable answer to who holds title; a buyer of a share needs a reliable answer to what claim the security represents. Such institutions economize on information by defining which facts matter for the transaction and which facts can remain outside it.

The same economizing process can also discard information that later becomes valuable. When production depends increasingly on intangible, cumulative, collaborative, and reusable inputs, the history excluded from the conventional asset record may contain information relevant to verification, bargaining, licensing, valuation, innovation, and dispute resolution. The economic problem is therefore not that markets remember possession and should instead remember everything. The problem is to determine when the information omitted by the possession record becomes sufficiently valuable that an additional institution of memory is justified.

II. Why Possession Is Cheaper to Record Than Contribution

Possession and contribution are different information problems. Possession can often be represented as an institutional state: A owns X; B holds 200 shares; C has assigned a license; D received payment. The record may be legally complex, but the state itself is comparatively compact. Contribution is a causal problem. It asks what input affected what outcome, how strongly, for how long, in interaction with which other inputs, under which counterfactual baseline, and with what degree of confidence.

Three differences matter economically.

- First, possession is ordinarily transaction-centered. The moment of transfer supplies a natural event around which institutions can organize a record. Contribution may precede measurable value by months or years and may continue to matter after the contributor leaves the organization.
- Second, possession is often discrete while contribution is frequently joint. An asset can have a current owner even when the underlying value emerged from hundreds or thousands of complementary inputs.
- Third, possession can be verified through recognized registries, contracts, and settlement records. Contribution frequently requires causal inference. The institution must distinguish correlation from material effect and direct inputs from interaction effects.

The resulting bias toward possession is therefore not necessarily evidence of institutional hostility to contributors. It is partly an outcome of comparative information costs. Markets preserve what can be standardized cheaply and what exchange requires them to preserve. The economic case for provenance must therefore show not only that contribution information is morally or historically interesting, but that the information has sufficient decision value to justify producing it.

III. Institutional Memory as an Economic Good

Institutional memory can be understood as durable, transferable information about economically relevant history. A provenance institution preserves evidence concerning contribution, context, temporal order, dependencies, performance, disputes, corrections, and later use. Its principal economic characteristic is that the information can survive the particular employment relationship, firm, contract, platform, or negotiation in which the contribution first arose.

The information is not itself the contribution, and it is not automatically a property right. It is an informational asset that can reduce uncertainty for later decision-makers. A prospective licensee may use it to evaluate an upstream asset. A worker may use it to establish a verified record of past outcomes when moving between firms. A firm may use it to identify high-value internal processes. A court or arbitrator may use it to narrow a dispute about creation or use. An investor may use it to distinguish an asset supported by a deep, verified development history from one supported chiefly by assertion.

The economic value of institutional memory is thus derivative. Memory is useful to the extent that a decision made with the information is better, cheaper, faster, more contestable, or less vulnerable to opportunism than the same decision made without it.

IV. Information Asymmetry: The First Market Failure

Contribution markets, where they exist at all, contain asymmetries on both sides. Contributors may possess private information about the quality, originality, difficulty, and causal importance of what they

created. Asset holders may possess private information about downstream deployment, revenue, integration, licensing, or reuse. Neither side necessarily observes the whole production-and-use chain.

Conventional proxies partly address this gap. Credentials signal expected capability. Reputation summarizes past observations. Employment history signals experience. Patents and copyrights establish defined legal facts. Firm brands reduce buyer uncertainty. Yet proxies can be lossy. They often describe the person, institution, or legal category rather than the particular contribution and its observable downstream performance.

A provenance record can reduce this loss when it links an identifiable contribution to verifiable context and outcome evidence. The potential gain is not that every buyer learns everything. The gain is a reduction in the cost of distinguishing stronger claims from weaker ones. Where that distinction affects contracting or investment, provenance has economic value as a quality signal. Where buyers already possess adequate information, the incremental value may be close to zero.

V. Transaction Costs and the Boundary of the Firm

Transaction-cost economics provides the clearest framework for deciding when contribution memory should become institutionalized. Markets are not free coordination mechanisms. Parties must search for counterparties, communicate information, bargain over terms, specify obligations, verify performance, monitor compliance, enforce agreements, and resolve disputes. When those costs are high, activity can be organized within firms or left untraded even though the underlying resource is productive.

Contribution information is expensive precisely because its relevant dimensions are difficult to specify in advance. Expertise may be asset-specific. Future applications may be unknown. Quality may be observable only after deployment. Attribution may depend on information controlled by the organization using the contribution. These characteristics increase the costs of contracting separately over each input.

A provenance institution changes this calculus only if it standardizes some previously bespoke tasks. Reusable identity attestations can reduce repeated verification. Standard contribution records can reduce search and due-diligence costs. Persistent performance evidence can reduce information asymmetry. Common dispute procedures can reduce enforcement costs. Portability can reduce the cost of re-establishing a contribution history with every new counterparty.

But provenance introduces costs of its own. It requires capture, validation, storage, privacy protection, model maintenance, attribution, contestation, correction, governance, and sometimes adjudication. The relevant economic comparison is therefore not provenance versus zero cost. It is the total transaction cost of the existing arrangement versus the total transaction cost of an arrangement that preserves more of the contribution history.

VI. The Net Institutional Value Test

The economic boundary of provenance can be stated as a simple decision rule. Let B_P denote the expected economic benefit generated by preserving contribution information and C_P denote the full institutional cost of doing so. Provenance is economically warranted when:

$$B_P > C_P$$

The benefit term should be interpreted broadly but not vaguely. It consists of measurable reductions in costs or improvements in decisions attributable to the information. A useful decomposition is:

$$B_P = \Delta I + \Delta V + \Delta K + \Delta B + \Delta D + \Delta M$$

where ΔI represents reduced information loss or adverse selection; ΔV reduced verification and due-diligence costs; ΔK reduced contracting and monitoring costs; ΔB improved bargaining or portability; ΔD reduced dispute and adjudication costs; and ΔM improved allocation, investment, licensing, or market matching.

The cost term can be decomposed as:

$$C_P = C_R + C_V + C_A + C_Q + C_G + C_{Pr} + C_{Ad}$$

where C_R is recording cost; C_V validation cost; C_A attribution and measurement cost; C_Q contestation and correction cost; C_G governance cost; C_Pr privacy and security cost; and C_Ad ongoing administration cost.

These expressions are an economic accounting framework, not an empirically calibrated formula. Their purpose is to force the theory to face its own cost ceiling. A provenance system that costs more to operate than the informational failures it corrects is not a successful application merely because its records are more complete.

VII. Incomplete Contracts: Memory Across Unforeseen Uses

Contracts are necessarily incomplete. Parties cannot specify every future state of the world, every use of an asset, every complementary contribution, or every technology that may later make a previously minor input valuable. The problem is pronounced for intangible contributions because reuse can be cheap, downstream applications can be distant from the original transaction, and value can emerge after the original parties have separated.

Institutional memory does not make a contract complete. It changes what can be established after an unforeseen state occurs. If the future use of a contribution was not contractible ex ante, a durable provenance record may nevertheless establish what was supplied, when it was supplied, the conditions surrounding the contribution, and how it entered the later system. That evidence may make renegotiation, licensing, adjudication, or voluntary recognition less costly.

This distinction is crucial. Provenance is not a retroactive contract. It does not convert every unforeseen use into a payment obligation. Its economic function is to preserve option value in information: when an unforeseen use becomes economically relevant, the historical evidence needed for a later decision has not disappeared.

VIII. Principal-Agent Problems and the Observability of Contribution

Organizations employ agents because decentralized knowledge and specialized action cannot be completely directed from the center. The same arrangement generates familiar monitoring problems. Principals cannot perfectly observe effort, agents cannot always demonstrate the causal value of effort, and both sides may possess private information about outcomes. Compensation systems therefore rely on proxies: hours, output counts, managerial evaluation, rank, team performance, or revenue measures.

Contribution memory can improve the information available to both sides when the economically important unit is not contemporaneous effort but a durable improvement. A process redesign, diagnostic annotation, software component, dataset, research method, or operating heuristic may continue creating value after the initial labor has ended. Conventional performance management may record that

the worker completed a project while failing to preserve the later dependency of downstream performance on the specific contribution.

The potential efficiency gain is better incentive alignment. If organizations can identify which contributions produce persistent gains, they can reward, retain, license, or redeploy those inputs more intelligently. The danger is equally clear: once a metric becomes consequential, participants have incentives to optimize the metric rather than the underlying value. Provenance therefore reduces principal-agent problems only when its measurements are harder to game than the proxies they replace and when uncertain attribution remains explicitly uncertain.

IX. Asset Specificity, Hold-Up, and Bargaining Power

Asset specificity creates a particularly strong case for institutional memory. A contribution may have large value inside one relationship but little observable value outside it. An engineer learns a proprietary production system and develops a refinement that works only with that system. A clinician contributes annotations inside one hospital network. A logistics specialist develops routing heuristics using one firm's historical data and infrastructure. Once the contribution is embedded, much of its value depends on complementary assets controlled by the organization.

This creates the possibility of hold-up. Before the investment, both sides may benefit from cooperation. After a relationship-specific contribution has been made, the party controlling deployment, records, customer access, complementary capital, or legal rights may hold the stronger bargaining position. The contributor may be unable to prove the contribution outside the firm, while the firm can continue using the organizational capability into which the contribution has been absorbed.

Portable provenance can alter the outside option without dictating the bargaining outcome. A contributor able to establish a credible record of what was contributed and what outcomes followed may carry more economically relevant information to another employer, licensee, investor, or adjudicator. The effect should not be assumed to favor contributors in every case. A verified record may also show that an asserted contribution had little measurable effect. The efficiency claim is narrower: bargaining based on better information is less dependent on whichever party happens to control the surviving records.

X. Contribution Decomposition: From History to Economic Information

A history of participation is not yet a history of economic contribution. If ten people worked on an outcome, a list of ten names cannot establish how the value depended on each person's input. Contribution decomposition is the analytical step that connects institutional memory to economic inference.

Let C denote a set of identifiable contributions and V(S) the value associated with a subset S of those contributions. The marginal contribution of a contribution c_i to a contribution set S can be represented as:

$$MC(c_i, S) = V(S) - V(S \setminus \{c_i\})$$

The expression is conceptually simple and empirically demanding. In complex systems one often cannot literally remove a historical contribution while holding everything else constant. A practical attribution system must therefore estimate marginal effects through perturbation, controlled comparisons, quasi-experimental designs, influence methods, or other domain-appropriate approaches. The resulting estimate is not the true contribution revealed by mathematics; it is an estimator whose quality depends on the value function, data, assumptions, baseline, and model.

The problem becomes more difficult because contributions interact. Two inputs can jointly create value that neither produces alone. A simple pairwise interaction term is:

$$\mathrm{Syn}(c_i, c_j) = V(\{c_i, c_j\}) - V(\{c_i\}) - V(\{c_j\})$$

Time also matters. Contributions can decay, persist, or propagate into later systems. An economically useful attribution framework therefore needs temporal persistence as well as contemporaneous marginal effect. Cooperative-game methods such as Shapley allocation offer one principled way to distribute a specified value function across contributors when the associated axioms are desired. They do not solve the prior empirical problem of constructing the correct value function, and computational approximations introduce additional error. Their role here is to show that institutional memory can, under suitable conditions, be transformed from a chronology into structured economic information without pretending that the transformation is exact.

XI. Path Dependence and Cumulative Innovation

The economy is path-dependent. Present products and capabilities inherit choices, discoveries, standards, routines, code, datasets, training, public infrastructure, and prior failures. Later contributors operate inside a possibility set partly created by earlier contributors. This makes the terminal asset a poor summary of the productive history even when it is a perfectly valid object of ownership.

Cumulative innovation creates two opposing risks. The first is erasure: early or infrastructural contributors disappear from the final asset record even when later work depends materially on what they created. The second is over-attribution: preserving a long ancestry of influences can encourage claims so diffuse that no transaction can close. A workable institution needs a materiality boundary.

The appropriate question is not whether a contribution appears somewhere in the ancestry. It is whether preserving information about that contribution changes an economically relevant decision enough to justify the cost. This connects path dependence directly back to the net institutional value test. History matters economically when forgetting it is costly.

XII. Positive Externalities and Unpriced Contribution

Some contributions generate benefits that the contributor cannot feasibly contract over. Knowledge can be reused by parties the contributor never meets. A method developed for one project can become a standard practice elsewhere. A dataset can support later research. An open technical contribution can improve products across firms. These are familiar externality conditions: social value and privately captured return can diverge.

Provenance does not eliminate the externality by itself. Recording who contributed what does not determine whether later users owe compensation. It can, however, make the externality observable. Once the origin and downstream dependency are visible, institutions can decide whether the appropriate response is no payment, reputation, licensing, prize mechanisms, contractual revenue sharing, public subsidy, a property rule, or another arrangement.

This separation is economically important. The information layer can be useful even where the optimal policy is not ownership. A society may want to measure spillovers in order to improve research funding or organizational incentives without converting every spillover into an exclusion right.

XIII. When Provenance Becomes Uneconomic

The strongest version of the institutional-memory argument includes the cases in which it should not be used. Provenance becomes uneconomic when the value of improved information is smaller than the cost of producing and governing that information.

Several conditions push toward that boundary:

- The contribution is too small relative to the cost of measurement.
- The causal relationship between input and outcome is too uncertain to affect a rational decision.
- The number of micro-contributions is so large that attribution costs overwhelm the allocable value.
- Existing contracts, reputational mechanisms, or organizational records already supply adequate information.
- The relevant value is predominantly tacit and cannot be represented with acceptable loss.
- Privacy or security protections necessary to preserve the record cost more than the expected coordination gain.
- Contestation and adjudication become more expensive than the disputes the system prevents.
- Metric gaming or strategic fragmentation of contribution creates new inefficiencies larger than the old ones.
- The persistence of records discourages experimentation, collaboration, or socially beneficial sharing.

These are not peripheral objections. They define the economic design problem. A provenance institution should be selective enough to remember economically material history and disciplined enough to forget what is not worth institutionalizing.

XIV. A Research Program: Does Memory Improve Markets?

The theory becomes economically meaningful only if it can be tested. The central empirical question is whether credible contribution histories improve observable outcomes relative to existing institutional arrangements. The most useful research program would compare domains, organizations, or transactions with different levels of provenance quality while avoiding the assumption that more detailed recording is inherently better.

The following hypotheses are testable:

H1 - Verification: Higher-quality provenance reduces verification and due-diligence time for transactions involving intangible contributions.

H2 - Contracting: Standardized contribution histories reduce contracting and renegotiation costs where future use is difficult to specify ex ante.

H3 - Mobility: Portable contribution records improve labor-market matching where conventional credentials are weak proxies for demonstrated performance.

H4 - Bargaining: Contribution evidence changes bargaining outcomes most strongly where asset specificity is high and one party otherwise controls the relevant records.

H5 - Licensing: Verified provenance increases licensing or commercialization of intangible contributions whose quality is otherwise difficult for buyers to assess.

H6 - Innovation: Attribution systems improve incentives for cumulative innovation only when measurement and claim-fragmentation costs remain below the resulting innovation gains.

H7 - Disputes: Credible audit histories reduce the frequency, duration, or cost of attribution and ownership disputes.

H8 - Investment: Investors assign different valuations or financing terms when intangible assets have deeper, independently verified provenance and performance histories.

H9 - Cost ceiling: Below a domain-specific value or materiality threshold, provenance lowers rather than raises net welfare because administration exceeds informational benefit.

The relevant outcome measures should be mundane enough to falsify the theory: hours of diligence, legal cost, time to agreement, bid-ask dispersion, licensing frequency, job-match quality, dispute duration, administrative expense, measured innovation output, or willingness to finance. If better provenance does not improve these outcomes after accounting for its cost, the economic case fails in that domain.

XV. Implications for the PBS/MPS Architecture

The economic analysis clarifies the proper role of the PBS/MPS architecture. Provenance should not be defended principally as a technological feature or as a moral claim to ownership. Its first economic task is informational: preserve economically material history at lower total cost than the opacity it replaces. Performance evidence then supplies observable information about outcomes. Contribution decomposition estimates how documented inputs relate to those outcomes. Only after those analytical steps does a governing legal, contractual, regulatory, or market institution decide whether an allocation, license, royalty, security, or other entitlement should follow.

This ordering protects the architecture from a common category error. If the ledger itself is treated as ownership, the framework assumes its conclusion. If the Performance Oracle is treated as perfect valuation, it ignores measurement error. If contribution decomposition is treated as exact causal truth, it ignores counterfactual and interaction uncertainty. If a Shapley-based allocation is treated as mandatory rather than conditional on the chosen value function and axioms, it converts a useful mathematical rule into an unjustified universal prescription.

The economically stronger proposition is more modest: better contribution information can expand the set of transactions and institutional choices that are feasible. Whether society should use that expanded possibility set for licensing, compensation, worker ownership, investment, governance, or no additional entitlement is a separate decision.

XVI. The Unequivocal Economic Claim

The paper can now state its central claim without requiring a theory of distributive justice.

Markets record possession better than contribution because possession is usually cheaper to standardize, verify, and transfer, while contribution is a costly causal history. When the economic value of that missing history exceeds the cost of preserving it, institutional memory becomes productive infrastructure.

This does not mean that every contribution should be recorded. It does not mean that every recorded contribution should be priced. It does not mean that every attributed contribution creates ownership. It means that institutional forgetting has an economic price, just as institutional remembering has an economic cost.

The relevant design objective is therefore neither maximal transparency nor maximal attribution. It is efficient memory: enough credible, portable, contestable history to improve decisions where the gains exceed the cost, and enough restraint to leave the rest outside the formal system.

Conclusion

The modern market economy is exceptionally good at preserving the state variables necessary for exchange. It knows who owns the parcel, who holds the share, which party signed the contract, what payment settled, and where the recognized right moved next. That institutional achievement should not be understated. It is one reason complex markets can function at scale.

But the same economy often stores the history of value creation inside temporary organizational records, reputations, memories, work products, and tacit knowledge. When those histories disappear, later markets inherit the asset without necessarily inheriting the information needed to evaluate the contributions that produced it. In a tangible economy this loss may frequently have been tolerable. In an economy increasingly organized around reusable knowledge, data, processes, creative inputs, and distributed expertise, the discarded information may itself become economically consequential.

The case for provenance is therefore a case about institutional economics before it is a case about entitlement. Preserve contribution history where doing so reduces total coordination costs, improves decisions, or opens efficient transactions that opacity prevents. Decline to preserve it where the measurement and governance burden is larger than the benefit. Treat attribution as an inference under uncertainty, not as a declaration. Keep evidence distinct from legal right.

The research question is finally empirical: when markets are given credible memory of contribution, do they allocate resources differently and better? If the answer is yes in identifiable domains, institutional memory is not merely a record of the past. It is economic infrastructure for the future.

Claim-Status Note

This paper is an author synthesis and proposed research framework. It does not claim that comprehensive contribution provenance is economically efficient in every domain, that contribution attribution is exact, that any particular attribution method is uniquely correct, that a provenance record creates a legal property right, or that PBS/MPS has production-scale empirical validation. The formulas presented here organize testable economic relationships and attribution concepts; implementation, causal validity, legal consequence, and net economic benefit remain domain-specific empirical questions.

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